DEMOCRACY, REDISTRIBUTION, AND INEQUALITY: EVIDENCE FROM THE ENGLISH POOR LAW

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Abstract
This paper tests whether inequality shapes the redistributive impact of democratization by examining changes to the governance of councils providing poor relief—rudimentary social insurance—in nineteenth-century Britain. An 1894 reform removed institutional features—a graduated franchise, property qualifications, the absence of a secret ballot, and the participation of unelected magistrates—that allowed landowners to control spending on poor relief after the 1832 Great Reform Act. The empirical analysis uses a new annual dataset of poor law spending from 1884–1905 to test whether higher pre-reform inequality amplified the effect of democratic reform on redistribution. The results support the Meltzer–Richard hypothesis: higher local income inequality led to higher poor relief spending after 1894. Areas where landed elites held political power saw smaller increases in expenditure, indicating that de facto elite influence muted the effect of democratization. These findings provide empirical support for models of democratization that focus on demands for redistribution. (JEL: D72, H11, N43)

Keywords: Democratization, Inequality, Redistribution, Meltzer–Richard, De Facto Power.

1. Introduction

The hypothesis that higher income inequality leads to greater government redistribution is extremely influential throughout the social sciences, but has received little empirical support. In the canonical model of Meltzer and Richard (1981), a poorer median voter demands greater spending as they face a relatively low tax burden. More recently, this insight has underpinned models of democratization (Acemoglu and Robinson 2000, 2001; Boix 2003). Yet testing these theories is complicated by the endogenous relationships between inequality, redistribution, macroeconomic factors, and political institutions: high inequality may, for example, make elites more capable of

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blocking post-democratization reforms, and more incentivized to do so (Acemoglu and Robinson 2006b). Consequently, there remains little causal evidence that inequality increases government spending and few studies have investigated whether and how inequality moderates the effects of democratic reform.[1]

This paper overcomes these endogeneity issues by exploiting democratic reforms imposed upon locally elected councils in nineteenth-century Britain. These local councils, known as boards of guardians, controlled spending on poor relief—the primary form of government redistribution before the modern welfare state. Elites benefited from four important institutional advantages: significant property qualifications, a graduated voting franchise, the absence of a secret ballot, and the presence of unelected landowners on councils. In 1894, all of these advantages were simultaneously removed by Parliament. I find that this exogenous shock to the political power of the poor led to greater increases in spending in areas of high income inequality, consistent with the Meltzer–Richard hypothesis. At the same time, I find that post-democratization spending also responded to local economic need and was reduced by elite de facto power. Government redistribution was thus shaped by a variety of factors beyond income inequality.

These findings provide clear empirical evidence for the widespread theoretical proposition that, under certain conditions, democracies redistribute more in the presence of greater income inequality. In particular, the paper studies a setting where, consistent with the assumptions of the Meltzer–Richard model, taxation was approximately proportional to income and spending was confined to redistribution. A significant literature has argued that those conditions rarely hold in modern-day political institutions (Moene and Wallerstein 2001; Iversen and Goplerud 2018). Yet those institutions are themselves endogenous, and likely shaped by elites, who may act to head off redistribution through either resisting democratization (Boix 2003), shaping a society's constitution (Acemoglu et al. 2012), or investing in de facto power (Acemoglu and Robinson 2006a). Such responses may explain the limited empirical support for the Meltzer–Richard hypothesis in previous studies, and hence point to income inequality as an important determinant of institutional development.

The paper begins, in Section 2, by presenting a conceptual framework that both clarifies the conditions under which the Meltzer–Richard hypothesis should hold, and distinguishes it from other theoretical channels through which inequality may affect spending. The hypothesis predicts specifically that inequality leads to higher redistribution by increasing the share of the tax burden borne by the rich. A simple model demonstrates that this prediction is more likely to hold when the political power of the poor is limited. Section 3 then explains that these conditions were plausibly met in nineteenth-century England: local councils spent money only on redistribution, raised revenue using a tax that was approximately proportional to income, and democratization remained incomplete. The setting thus offers many of

1. Results regarding how inequality affects government redistribution in general have been somewhat mixed, with observational studies finding a mixture of positive and negative results—see De Mello and Tiongson (2006, Table 1) for a summary.

the advantages of testing the Meltzer–Richard model in the laboratory (Agranov and Palfrey 2015), while incorporating the complexities of a real-world setting.

I test how local inequality shaped reactions to democratic reform using a new annual dataset of spending on poor relief between 1884 and 1905. As explained in Section 5, I use the 1894 democratic reform to poor law governance as an exogenously imposed "treatment" that affected all poor law unions. The effect of inequality on spending per capita is then identified by including interactions between pre-reform inequality and an indicator variable for the post-reform period, and controlling for both time-invariant and time-varying factors. To test the Meltzer–Richard mechanism, I estimate income inequality as the ratio of mean rental value to the median wage in each poor law union. Because local taxation was based on rental values, this measure directly corresponds to the relative tax burden of the rich versus the poor—the channel through which inequality affects demand for redistribution in the Meltzer–Richard model. Further, I account for other dimensions of rural inequality, allowing me to disentangle the Meltzer–Richard mechanism from other competing channels.

The results, presented in Section 6, provide robust support for the Meltzer–Richard hypothesis. Democratic reform was followed by greater increases in spending in areas with higher income inequality. The relationship holds when controlling for demographic characteristics affecting the demand for poor relief, financial factors constraining spending, and for union-specific time trends. Further, the estimated effect of income inequality remains strong after controlling for measures of land inequality, economic need, and the power of local elites. Overall, these findings are consistent with inequality affecting spending through the relative tax burden on rich and poor, as predicted by Meltzer–Richard, rather than through alternative channels.

Three additional findings emerge from the empirical analysis. First, the effect of income inequality was particularly pronounced for the provision of out-relief—the most clearly redistributive aspect of poor law policy—consistent with the Meltzer–Richard mechanism. Second, post-reform spending was lower in areas in which local elites held political power before 1894, consistent with them retaining de facto power after reform (Acemoglu and Robinson 2006a). This effect is most evident for less publicly-salient items of expenditure, suggesting that elites could blunt the impact of democratization when the poor were unable to easily scrutinize policy. Third, spending increased by more in areas with arable agriculture, suggesting that democratization responded to economic need.

Several studies find that democratization increases government expenditure at both national and local level, but few examine whether inequality moderates the effect. Consistent with the idea that newly enfranchised voters push for redistributive expenditure, prior work generally finds positive effects of democratization on both tax revenue (Aidt et al. 2006), and social expenditure (Lindert 1994, 2004).[2] The focus on redistributive expenditure in this paper broadens the small literature investigating the

2. Reforms such as the 1894 Local Government Act analyzed in this paper help explain how expansions of the national franchise, emphasized by Lindert (2004), translated into increased redistribution by local governments. Studies of tax incidence have been less supportive of the link between democratization and

effects of democratization within a country, which has mostly focused on expenditure on public goods or education (Husted and Kenny 1997; Aidt et al. 2010; Cascio and Washington 2014; Fujiwara 2015; Chapman 2018, 2024). Similarly to the findings here, Chacon and Jensen (2020) find that greater elite de facto power led to lower taxation in the U.S. following the enfranchisement of black voters during Reconstruction, without testing the role of income inequality.

Two recent papers have investigated how heterogeneity in inequality moderates the effects of democratization at national level. Acemoglu et al. (2015) find mixed results, but suggest that higher inequality may be associated with elite capture and hence lower taxation post-democratization. Dorsch and Maarek (2019) find that post-democratization income inequality dynamics are affected by initial inequality, but fiscal redistribution does not appear to be an important channel. Both sets of results are consistent with the findings in this paper that local elites were able to repress redistributive spending post-reform. However, the analysis here provides a more nuanced story in which elites could only reduce less publicly-salient forms of expenditure. Moreover, I observe a separate effect through which income inequality positively affected spending. The difference in findings may reflect the endogeneity problem inherent in national-level studies—elites may only democratize when they can limit the effects on redistribution. The exogenous democratic reform exploited in this paper avoids such complications.

This paper also contributes to the large literature on the history of the English Poor Law. Much of that literature has focused on understanding the effects of the 1834 New Poor Law (Boyer 2006; Melander and Miotto 2023), and the continuities between the Old and New Poor Law (Digby 1978). A handful of papers have quantitatively analyzed aspects of the operation of the poor law after 1850 (MacKinnon 1986, 1987; Brueckner 2023), but do not focus on the governance of poor relief. In contrast, this paper investigates the democratization of poor law unions in the 1890s and demonstrates how poor relief policy was shaped by the interaction between local political institutions and economic structure.

The paper concludes by considering the implications of these results for theories of democratization, and placing the findings in the broader context of British political history—a motivating example for many of those theories. The empirical analysis suggests that decentralization allowed rural elites to limit the extent of redistribution long after Britain's 1832 Great Reform Act—an example of "elite-biased" democracy (Albertus and Menaldo 2014). By taking control of the institutions established by the 1834 New Poor Law, landowners were able to escape the auspices of the increasingly democratic Westminster Parliament; it was not until agricultural laborers obtained the Parliamentary franchise that rural local governments were democratized. Maintaining control of poor law institutions thus enabled elites to reduce the cost of democratization for a further sixty years.

redistribution (Aidt and Jensen 2009a,b; Scheve and Stasavage 2010, 2012). See Acemoglu et al. (2015, Section 21.3) for a detailed review of the literature.

2. Conceptual Framework

This section introduces a simple theoretical framework that informs the empirical analysis. The framework identifies conditions under which income inequality may shape the effect of democratization on redistribution, and also identifies other factors that may confound this relationship. Here I sketch the main insights of the model, with a full formal treatment provided in Appendix A. Section 3 then describes how the historical setting fits the theoretical assumptions, and Section 4 explains how the hypotheses are operationalized.

Consider a population normalized to size 1, consisting of poor (P, share $\lambda > 0.5$) and rich (R, share $1-\lambda$) citizens. The government provides poor relief r funded by a linear tax, t, on income. Rich citizens always receive an income and never receive relief. Poor citizens are employed and receive income with probability π, and always receive at least partial relief. All agents have concave utility over individual consumption, $u(\cdot)$, with $u'(0) \to \infty$. I also assume, following Moene and Wallerstein (2001) that $u(c)$ exhibits constant relative risk aversion (CRRA), with $\gamma = -c(u''/u') > 1$.[3]

Utilities of each group are then given by:

$$U_R = u\left(y_R(1-t)\right)$$
$$U_P = \pi u\left(y_P(1-t) + \psi r\right) + (1-\pi)u(r) + \kappa t,$$

where y_P and y_R are incomes of the relevant group if employed. $\psi > 0$ represents the value of relief received by employed poor citizens, either through income support or payments to dependents. If $\pi < 1$ and $\psi > 0$, then government spending has two effects. First, it redistributes from rich to poor. Second, it provides insurance against negative income shocks, such as unemployment. $\kappa \geq 0$ captures instrumental utility from taxing the rich due to fairness concerns, inequality aversion, or class conflict.

The amount of relief provided in equilibrium is determined by maximizing the following social welfare function, subject to a balanced budget constraint:

$$\max_r W = \varphi U_P + (1-\varphi)U_R$$

such that $r\lambda(\pi\psi + (1-\pi)) = (t - D(t))\bar{y}$. $D(t)$ is a strictly convex function representing administrative costs and potential tax base erosion due to higher taxation.[4] φ reflects the political weight placed on the welfare of the poor in the institution determining the level of relief.[5] Because the poor constitute a numerical majority

3. Appendix A presents the model without the CRRA restriction. The condition $\gamma > 1$ underpins the result (here and in Moene and Wallerstein (2001)) that higher inequality may not lead to higher government spending if relief is targeted at the unemployed.

4. Specifically, similarly to Acemoglu and Robinson (2006b), I assume that $D' > 0$, $D'' > 0$, $D(0) = D'(0) = 0$, and $D'(1) = 1$.

5. In a standard utilitarian welfare function, φ would embed the share of citizens in each group.

($\lambda > 0.5$), the extremes of $\varphi = 0$ and $\varphi = 1$ are equivalent to median voter models with voting rights restricted to the rich or extended to all citizens, respectively.

This framework is sufficient to reproduce the central insights of the Meltzer–Richard model. Setting $\pi = \psi = 1$ and $\kappa = 0$, and interpreting $D(t)$ as labor-supply distortion from taxation, essentially reduces the model to the canonical Meltzer–Richard (1981) setup: a linear tax financing a universal lump-sum transfer chosen by the median voter. This leads to the standard prediction that a higher mean–median income ratio raises redistribution. Specifically, higher inequality lowers the poor's relative tax burden from providing relief and so their preferred level of relief increases.

The model also illustrates how the Meltzer–Richard predictions rely on a number of strong assumptions. The tax rate is linear, meaning that the poor always share the tax burden. Further, the direction of the relationship between inequality and the size of government is dependent on the nature of government spending. If spending purely provides insurance—$\psi = 0, \pi < 1$—then greater inequality leads poor voters to desire lower government spending. This latter finding is a key insight of Moene and Wallerstein (2001). The model also abstracts from other potential factors mentioned in the literature that may mean that income inequality does not lead to higher redistribution—such as the potential for middle class control of government, multi-dimensional policy space, or party alignments.[6] A core contribution of this paper is to identify a setting where these assumptions plausibly hold, meaning that the Meltzer–Richard hypothesis can be tested empirically.

A distinctive feature of this framework is the role of democratization, captured by the political weight parameter φ. Notably, the standard Meltzer–Richard prediction—that higher inequality increases government spending—is less likely to hold when the poor have substantial political weight. Specifically, when φ is high, a mean-preserving spread (an increase in y_R/y_P while keeping $\bar{y}$ fixed) raises spending only if the redistributive, rather than the insurance, motive for government spending is sufficiently strong. In contrast, if the poor's political weight is limited (φ is low), then higher inequality increases spending without these conditions.[7] Intuitively, weaker democracy reduces the weight of those citizens—the employed poor—for whom higher inequality may translate into lower desired government spending. A mean-preserving spread involves a reduction in y_P and an increase in y_R. This reduces the demand for the insurance element of government spending among the poor, but also weakens the opposition to higher expenditure among the wealthy because their marginal utility cost of higher taxation is lower.[8] When φ is low, the preferences of the rich dominate.

6. See the review by Acemoglu et al. (2015) for a discussion of these possible factors.

7. More precisely, there exists $\bar{\varphi} \in (0, 1]$, such that an increase in inequality strictly increases equilibrium government spending if $\varphi \in (0, \bar{\varphi})$. Government spending is also increasing in income inequality if employed citizens are marginal beneficiaries of more relief or, as in Moene and Wallerstein (2001), when risk aversion is not too strong. See Proposition 1 in Appendix A for details.

8. Note that Moene and Wallerstein (2001)'s median voter framework captures the effects of the reduction in the income of the poor, but excludes any counterbalancing effect of increasing the income of the rich.

The framework also highlights other channels that can confound empirical tests of the Meltzer–Richard predictions. $\varphi \in (0,1)$ reflects the scenario in which the poor have some political power, but are constrained by institutional factors. In models of democratization, such as Acemoglu and Robinson (2006b), φ is endogenous to inequality: the wealthy may grant less political power to the poor in unequal societies, or maintain *de facto* influence even after losing *de jure* control (Acemoglu and Robinson 2006a). The employment probability, π, reflects the need for social insurance, while ψ reflects the extent to which relief reaches employed households through, for instance, provision to dependent family members. κ captures motives for redistribution not tied to consumption, such as resentment toward the rich. Each of these parameters represents a factor that may be correlated with income inequality, and that can affect government spending, without capturing the Meltzer–Richard mechanism.

In summary, the model identifies conditions under which income inequality—measured by the mean-to-median income ratio—should increase government expenditure, as in Meltzer and Richard (1981). The empirical analysis tests this prediction in a setting where key theoretical assumptions were plausibly satisfied: taxation was roughly proportional and fell partly on the poor, government spending benefited a broad swathe of households, and political institutions were imperfectly democratic. I also test alternative channels, allowing me to distinguish the Meltzer–Richard mechanism from other channels—such as class resentment, elite capture, or economic need—through which inequality may affect government spending.

3. Institutional Context

To test the effect of inequality on government spending, I exploit the fact that redistributive government expenditure in nineteenth-century England was provided by autonomous, locally elected councils. Poor relief served as the main safety net against poverty in Britain for several centuries, providing rudimentary social insurance against a range of economic shocks. This section first provides a general overview of the New Poor Law. The second subsection argues that the median voter would likely benefit from higher poor relief expenditure. The third subsection discusses the politics of the poor law and the democratic reform exploited in the empirical analysis.

3.1. The New Poor Law

Under the infamous 1834 “New Poor Law,” poor relief was controlled by a set of approximately 600 autonomous local authorities—“Boards of Guardians of the Poor”. These councils determined both the level and type of relief within their respective “Poor Law Unions,” raising funds through local taxation. While the guardians administered poor relief, they did not have responsibility for spending on infrastructure, local public services (provided by town councils), or education (controlled by local school boards).[9]

Spending on poor relief decreased after the introduction of the New Poor Law and did not recover until the 1890s, as shown in Figure 1. The newly-established boards of guardians significantly decreased expenditures after 1834 (Boyer 2018, Table 2.1), with per capita spending falling approximately 30% by 1860. Spending then remained at a similar level—fluctuating in response to economic crisis—for the next thirty years. Importantly for our purposes, there was little growth in spending in the ten years prior to democratic reform but a sharp increase immediately afterwards.

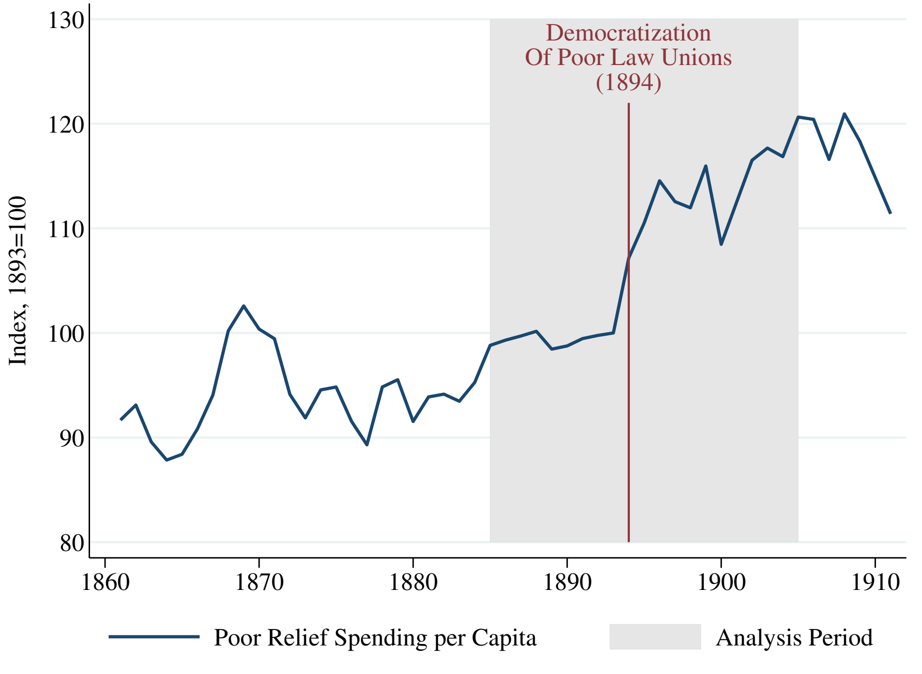

FIGURE 1. Poor law spending increased immediately after democratic reform in 1894. The figure displays annual average real spending per capita across the 208 poor law unions included in the regression sample. A supremum Wald test estimates that a structural break in poor relief spending per capita occurred in 1894. The null hypothesis of no structural break is rejected at a 1% significance level. See Appendix Table B.2 for pre- and post-reform annual growth in spending by census division.

The most contentious aspect of the New Poor Law was the attempt to force recipients to enter the workhouse—the infamous "workhouse test"—rather than receive relief outdoors ("out-relief"). Guardians were prohibited from offering outdoor relief to able-bodied men and, in the 1870s, vigorous efforts were made to prevent

9. The guardians sometimes held other roles that involved spending on public goods—see Appendix C.3.

any pauper receiving support outside a workhouse—a policy known as the "Crusade Against Outrelief". The number of outdoor paupers fell by more than a third on average, causing considerable resentment as elites were seen to be abandoning their paternalistic role in the community (Hurren 2015).

Guardians held a great deal of autonomy in administering poor relief, including in the implementation of the Crusade, as we can see in Figure 2. Guardians could determine who was eligible for relief, the generosity of payments, and whether support should be monetary or in-kind. Consequently there was considerable variation across unions in the generosity of spending and the repression of outdoor relief (bottom-left panel).

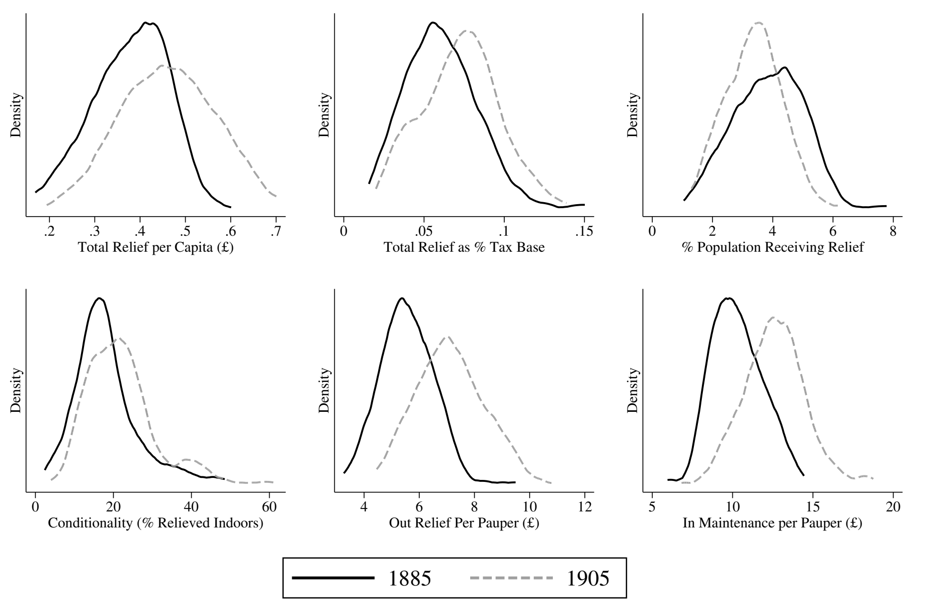

FIGURE 2. Poor law policy varied considerably across poor law unions. The figure displays kernel density for 208 poor law unions included in the regression analysis. Three poor law unions with 1905 in-maintenance per pauper of over £20 are excluded for display purposes.

Guardians had to finance their spending locally, predominantly through a proportional tax on the rental value of occupied property. Taxes had to be levied on owners and occupiers of land and buildings, meaning that all households, in principle, shared the cost of relieving the poor. The guardians decided the level of taxation, but had no ability to alter this tax structure or tax other sources of income such as profits. The tax rate was thus approximately proportional to income, and hence consistent with the framework in Meltzer and Richard (1981) and many other political economy models. The reliance on property taxation also implies that the local tax base was relatively inelastic in the short run, meaning that higher taxes were unlikely to have significantly eroded the tax base.

Other features of the historical context suggest that the assumptions of the Meltzer–Richard model are likely to hold in this setting. Contemporary debates surrounding the Poor Law focused on normative questions regarding the appropriate level and form of relief, rather than on efficiency concerns, such as potential disincentive effects of higher taxation. Unions did not have to consider whether more generous benefits would stimulate in-migration, both because migrants would not automatically be eligible for relief, and because the levels of relief were relatively small.[10] The focus of poor relief on relieving the destitute, and the accompanying stigma, meant that transfers could not be directed towards a middle class as in the "Modified Director's Law" introduced by Acemoglu et al. (2015). Poor law politics was not affected by social cleavages, such as ethnic fractionalization, that could undercut support for redistribution among the poor (Alesina et al. 1999). Collectively, these institutional features provide a setting that is well aligned with the Meltzer–Richard framework, and that avoids several possible countervailing influences of inequality on government spending.

3.2. Recipients of Poor Relief

The poor law was the most important safety net against economic shocks, and many citizens were at risk of relying on relief each year. The proportion of the population receiving relief in the 1870s was generally around 6%, but could rise above 20%.[11] Relatively few paupers were able to work: one-third of paupers were children, and most adult paupers were classified as "non-able-bodied," a category which included the temporarily sick, the mentally ill, the permanently disabled, pregnant women, and the elderly. By the 1890s much poor relief was targeted at the elderly—in 1892 30% of those over 60 received relief (Boyer and Schmidle 2009). Further, these statistics likely underestimate the number of those requiring support, as dislike of the workhouse deterred many eligible individuals from even applying. MacKinnon (1986) suggests that around 10–20% of the population were sufficiently poor to be at real risk of requiring *indoor* relief, much lower than the proportion seeking relief at all.

More generous poor relief policy also benefited citizens indirectly, by removing the need to financially support their relatives. Apart from moral obligation—amplified by the fear of the workhouse—individuals could be legally forced to financially support their relatives before relief was granted, or to repay the cost of relief ex post (King 2000, p20). Moreover, the concept of "relative" was sufficiently broad that even friends could be called upon to pay for the support of the elderly (Thomson 1984). Fewer restrictions

10. The poor law was still governed at this time by the "law of settlement" that meant residence was necessary to be eligible for relief. By the late nineteenth century the practical significance of these laws was beginning to fade, but still provided clear legal grounds to refuse to provide relief.

11. The "stock" of paupers was reported on two days each year; the proportion receiving support over a year is estimated using estimates from MacKinnon (1988) that the 1892 ratio of paupers receiving relief over the year to those relieved on January 1 was 2.24. Appendix C.4 includes a detailed classification of paupers.

on who received outrelief thus allowed citizens to pass the burden of supporting the destitute to the community.

Reliance on poor relief was particularly high in the rural poor law unions studied in this paper. Poverty in this period was extremely high—Gazeley and Newell (2011) estimate that nutrition levels were lower than in 1980s rural India—and agricultural laborers were particularly vulnerable, being unable to "make ends meet at all if it were not for charitable gifts—sometimes of coal, sometimes of food or clothing" (Rowntree and Kendall 1913). In the poor law unions I study, an average of 59% of male occupations were agricultural laborers, implying that the typical household was likely very poor.[12] Not only were wages low, the seasonal nature of agriculture provided an additional need for short-term income protection. As such, it is likely that the median voter in these areas would directly benefit from more generous poor relief policy.

3.3. The Democratization of Poor Law Institutions

Under the 1834 New Poor Law boards of guardians were elected, but elites held four institutional protections which allowed them to maintain control over spending.[13] First, elected guardians were supplemented by unelected *ex officio* guardians, including all magistrates—typically local landowners—residing in the poor law union. Second, property qualifications prevented poorer citizens from standing as poor law guardians. Third, guardians were elected under a graduated franchise under which the wealthy held multiple votes. In principle all property occupiers—essentially heads of household that owned or rented property—could vote unless they had received relief in the previous year. However, the number of votes varied depending on the value of the property owned or occupied: a single voter could receive up to twelve votes. Finally, there was no secret ballot in place. These provisions were openly acknowledged as protections for landholders (House of Commons 1878, paras 864–866, 5076). This undemocratic electoral system remained in place for sixty years, until the wide-ranging 1894 Local Government Act ("LGA") removed all four protections for the elite.

The LGA marked the culmination of two long-running movements in Victorian legislation—the rationalization of the system of local government, and democratization of government at all levels. Britain's national institutions had been democratized through the three Reform Acts of 1832, 1867, and 1884. In each case, national reform was followed by reforms to local governments: municipal boroughs in 1835, 1869, and 1878, and then the establishment of county councils in 1888. The primary purpose of the 1894 Act was to complete the task of local government reform in rural areas through establishing a system of elected parish and district councils across England

12. Further, analysis of the 1881 100% census sample shows that the median head of household's occupation, according to the Hiscam occupational score metric (Lambert et al. 2013), was in farming in 89% of unions. In the remaining 11% of unions the score was at a similar level.

13. See Appendix C.1 for a detailed description of the election system.

and Wales. Reform of the poor law was a secondary consideration, but once the boards of guardians were included in the provisions of the Act even those MPs concerned that democratization would lead to higher expenditure struggled to defend the status quo as "when the franchise in their other institutions was democratic it was quite impossible much longer to keep the Poor Law system on a different basis" (House of Commons 1893b, para 91). The decision to democratize the boards thus resulted from a reluctant acceptance that the logic of democracy precluded leaving patently undemocratic structures in place (Hurren 2015, p.215).

To understand how the LGA affected local politics I analyze the evidence provided to the 1909 Royal Commission on the Poor Laws—Appendix Table C.1 summarizes the evidence for each of thirty unions, and Appendix C.2 provides further discussion—combined with historical literature. Delving into local detail is necessary because rural poor law politics was predominantly local, with national parties holding little presence. The content and nature of political debate consequently varied considerably across the country, reflecting both the needs of local communities and social relations between landlord and tenant.

This analysis indicates that the LGA led to a marked shift in political power towards the poor. After 1894 guardians had to account for the views of the poor, since "guardians were not now nominees of large ratepayers but men...who know the wants and miseries of the poor" (Channing 1918, cited in Hurren, 2015, p.239). As usual in nineteenth-century British politics, few poor law elections were contested, but the perception that relief had been denied unfairly could lead to protests and trigger electoral contests—often with high turnout. In some areas, local agricultural unions provided organizations through which demands could be formulated, even in the absence of political parties. Further, "the publicity given to [the guardians'] work by means of newspaper reports [made] them even more liable to undue pressure from their constituents"(House of Commons 1909, p.101). Consequently the 1894 democratic reforms led to significant changes in the composition of boards of guardians, enabling, at least in some cases, dramatic changes in policy. In the words of an assistant commissioner to the Local Government Board: "It was complete revolution...some boards of guardians, certainly for a time, made radical changes and in some cases gave outdoor relief in the most lavish way" (House of Commons 1909, Appendix I, para 2045).

Nevertheless, the poor continued to face political barriers after 1894. Despite the removal of property qualifications, poorer citizens did not have the time or the financial resources to sit as guardians. While landowners were largely removed from boards, it was farmers that tended to control decision-making as "the country gentleman is unable to secure election, or holds aloof because he is not willing to seek election, and on the other hand, the country laborer is missing because he cannot afford the time" (House of Commons 1909, p.105). Malapportionment of electoral districts within Poor Law unions could prevent even broad popular support from translating into control of the board of guardians. Consequently, sharp spending increases post-reform tended to be short-lived. Thus, while the LGA increased the political influence of the poor, it did not allow them to fully control poor law policy. In fact, the empirical analysis provides evidence that the elite were able to reduce less politically salient spending.

Hurren (2015) provides a detailed history of the Brixworth union that highlights the political dynamics at play. Brixworth, controlled by the wealthy and politically-influential Earl Spencer, was a "model" union for the Crusade against Outrelief, cutting both the number of outdoor paupers and outdoor spending by over 95% between 1871 and 1894. To achieve these cuts, guardians prosecuted the adult children of elderly paupers for the costs of relief, refused relief to those receiving charity, reduced the number of relieving officers and dealt with claims directly, and referred elderly applicants to private charity. After the Third Reform Act, rural areas became increasingly politicized, leading to the 1893 foundation of a pressure group demanding the reintroduction of outrelief. Prior to 1894, however, they were unable to gain control over the board of guardians—the LGA enabled them to do so, and consequently to both reintroduce outrelief and upgrade the union workhouse.

In summary, poor law unions were autonomous political units, with control over spending on poor relief. Boards of guardians were elected locally, with contests centred around local issues that were of relevance to a large swathe of the population. The 1894 Local Government Act swept away the remaining vestiges of protection for landed elites. In the following sections I analyze the consequences of that democratic reform.

4. Data

This section summarizes the data used in the empirical analysis. I first discuss the sample of unions analyzed, and then the datasets used.[14]

4.1. Sample

The paper focuses on rural poor law unions for both conceptual and practical reasons. Conceptually, a focus on rural areas means that we compare unions that are similar in economic structure, but differ in inequality. Specifically, these unions represent settings with a small landed elite and a homogeneous mass of agricultural workers, as in the model of Acemoglu and Robinson (2000). Further, as discussed in the previous section, the median voter likely viewed poor relief as an important safety net. Practically, the main measure of inequality (the mean–median ratio) relies on the agricultural wage, which is only appropriate in rural areas. Restricting the sample to unions that were rural in character throughout the period also ensures that changing poor relief expenditure does not reflect trends in urbanization.

Rural poor law unions are identified using a classification as "rural or mainly rural" by the 1909 Royal Commission on the Poor Laws. This classification was undertaken

14. Full details and summary statistics are provided in Appendix B. The analysis combines newly collected data with several existing datasets, including Southall et al. (1998, 2004, 2023); Schürer and Higgs (2014); Reid et al. (2018) and Chapman (2022, 2023).

long after the LGA, ensuring stability through the analysis period. I remove poor law unions that underwent substantial boundary changes during the period: those that were established or abolished between 1860 and 1905, and those where the cumulative change in boundaries over the same period exceeded 15% of the 1881 population. Seventeen unions were excluded as the income inequality measure cannot be calculated due to missing wage data.[15] The main regression sample then consists of 208 poor law unions.

4.2. Data Sources

The empirical analysis uses two new datasets. The first measures union-level poor law activity using parliamentary papers. The second identifies the location and characteristics of elites across England and Wales. These data are then combined with information on local demographic structure and geographic characteristics.

Poor Law Data. This dataset includes annual information on unions' spending, revenue sources, tax base, and the number of paupers relieved in each year 1860–1913. I use these data to construct my main dependent variable—the total spending on poor relief per capita—and several control variables. An 1893 parliamentary paper identifies whether the pre-reform chairman of each union was elected or appointed ex officio (House of Commons 1893a).

Local Elites. I geolocate the residences and landholdings of rural elites (gentry and aristocracy) in each union, using information from Bateman (1883) and Walford (1886). This process allows me to identify the presence of peers of the Realm and the very wealthy in each union, and also to estimate the share of land owned by these elites.

Other Union Characteristics. Demographic and occupational characteristics are obtained using census data, and are interpolated geometrically to construct annual series for use in the panel regressions. The suitability of land for cereal agriculture in each union is estimated using data from the Global AgroEcological Zones project of the Food and Agriculture Organization (FAO/IIASA 2011) and 1881 boundary maps. Annual county-level data on the percentage of land devoted to different crops was collected from the *Annual Agricultural Returns*.

5. Empirical Approach

This section first outlines the empirical framework used to investigate the relationship between income inequality and the effects of democratic reform, and then introduces the inequality measures used to operationalize the theoretical hypotheses from Section 2.

15. Data was missing for three counties—Huntingdonshire, Rutland, and Suffolk. These missing values do not appear to bias the results: coefficients for other measures of inequality are similar when using the full sample ($N = 225$) and the smaller sample ($N = 208$) used in the main analysis.

5.1. Empirical Specification

I test whether the impact of the 1894 Local Government Act varied with the level of pre-reform inequality. Critically, democratization was imposed on local councils, avoiding endogeneity issues inherent in national-level democratizations. While national elites shaped the LGA, the democratic reforms it imposed were exogenous to each poor law union, allowing the effect of inequality on post-reform spending to be causally identified.

Specifically, I estimate:

$$y_{it} = \beta \text{inequality}_i \times \text{post1894}_t + \gamma_0 X'_{it} + T_t + \alpha_i + \varepsilon_{it} \tag{1}$$

where i indexes poor law unions and t indexes years. y is poor relief spending per capita. X is a vector of control variables, T represents year fixed effects, and α represents poor law union fixed effects. ε is an error term. Union fixed effects control for time-invariant differences in the level of relief in each union (e.g., culture). Year fixed effects capture events common to all unions, such as national economic shocks. As the level of inequality is measured at a single point in time pre-reform, the level variable is absorbed by the union-fixed effects. The measures of inequality are introduced in the following subsection.

The main coefficient of interest is β, which relates to the interaction between pre-reform inequality and the post-reform period.[16] For estimates of β to be interpreted as capturing a causal effect of the interaction between inequality and democratic reform, two key identifying assumptions must hold. First, that in the absence of democratic reform, and after adjusting for control variables, there would have been no relationship between inequality and changes in poor law spending. This assumption is similar to the "conditional parallel trends" assumption in a difference-in-difference framework. Second, any reaction to the 1894 democratic reform must reflect a response to local inequality rather than another characteristic correlated with inequality. Both assumptions require careful justification given that the level of inequality is not a randomly-assigned treatment and is, in particular, strongly correlated with the type of agriculture—a point discussed in detail in Section 5.3 below.

I use four approaches to address the concern that the level of inequality may be correlated with underlying trends in poor law spending. First, I include a large set of control variables including (i) a vector of "crop controls" capturing county-level land usage to account for differences in agricultural type; (ii) "demographic controls" to account for differences in demand for poor relief, including population density, population, age structure, and decadal variation in the number of paupers per capita; and (iii) "financial controls", including the log tax base per acre and the percentage of revenue from taxation, to account for differences in ability to fund poor relief. Second, the results remain similar when incorporating union-specific linear trends,

16. To use terminology more common in political science, inequality acts as a "moderator" of the reform's effect (see, for instance, Hainmueller et al. 2019).

accounting for any (linear) differences in the trend of poor law spending before 1894. Third, following Gentzkow (2006), I flexibly control for differences in the time path of poor law spending driven by the endogeneity of inequality by including interactions between county-level observable characteristics in a base year and quartic time trends. Finally, I test whether union-level inequality is associated with pre-trends in either local spending or other observable characteristics—there is little evidence of either, as shown in Figure 4. Consequently, it does not appear that poor relief would have evolved differently in more unequal unions had the 1894 reform not occurred.

To address the second concern—that the estimated β could capture the effect of some other variable correlated with inequality—I allow for heterogeneous reactions to the democratic reform according to other district characteristics. In particular, I allow for interactions between the post-1894 dummy and various measures of economic need, pre-reform economic decline, the ease of out-migration, characteristics of urban surroundings, and the importance of seasonal agriculture. The main findings are robust to including these controls, providing further evidence that the results capture a causal effect of inequality on the impact of democratic reform.

The causal interpretation of the results would be threatened if the broader changes enacted by the LGA affected poor relief through channels other than the democratization of boards of guardians. The creation of parish councils poses little concern, as they had limited spending or tax-raising powers and failed to inspire voter interest (Keith-Lucas 1952, p.42). However, the establishment of rural district councils is more problematic because guardians also served as representatives on these councils. Although the bodies were organizationally distinct, this overlap means that guardians could plausibly have been elected based on considerations other than poor relief. Further, the LGA expanded the range of spending controlled by these bodies, meaning the guardians' role as rural district councillors changed alongside electoral reform. Appendix C.3 discusses this potential issue in depth and demonstrates that these other changes are not a major concern in practice.

One final complication is that the 1894 reforms were likely, at least to some extent, anticipated. The bill itself was introduced in the House of Commons in March 1893, more than eighteen months before the first elections under the new electoral system. Further, in late 1892, an intermediate step of lowering (but not removing) property qualifications for elected guardians was taken by the Local Government Board. This change likely had a limited direct effect on policy, both because of the remaining institutional protections for landowners and because only some of the guardians on the boards were elected each year. However, it is possible that guardians adjusted their policy to increase the chances of winning elections later. Such a response would mean that the effects of reform are underestimated.

5.2. Measures of Inequality

The key challenge for the empirical analysis is to disentangle the effects of different aspects of inequality, and in particular isolate the channel highlighted by Meltzer–Richard. To do so, I construct several variables capturing different dimensions of inequality. To test the Meltzer–Richard hypothesis I construct a measure of income inequality that is directly related to the relative tax burden of the rich and poor. Further measures capture possible fairness concerns, elite de facto power, and the presence of economically vulnerable groups. Including these measures together in the same regression provides a robust test of whether inequality impacts government spending through fiscal incidence, as predicted by Meltzer–Richard, or through alternative channels.

TABLE 1. Operationalisation of channels linking inequality to government spending.

Channel	Predicted Effect post-Democratization	Empirical Measures
Relative tax burden / Meltzer–Richard ($\bar{y}/y_P$)	Income Inequality $\longrightarrow g \uparrow$	Mean rental cost/ Median wage
Elite De Facto power ($1-\varphi$)	Elite power $\longrightarrow g \downarrow$	Presence of a peer Unelected chairman
Inequality aversion / Fairness concerns (κ)	Salient class differences $\longrightarrow g \uparrow$	% Land owned by economic elite % Large farms Cuts to outdoor relief 1866–85
Economic need (π, λ)	Need $\longrightarrow g \uparrow\downarrow^{\dagger}$	Cereal suitability index % Agricultural laborers % Over 64

Notes: g is the spending on poor relief per capita. † Theoretically, the prediction is ambiguous: greater economic need mechanically increases total spending due to a higher number of recipients, but also raises the marginal cost of relief, potentially reducing the optimal level of relief provided.

Table 1 connects each measure to the conceptual framework presented in Section 2, and summarizes the predicted relationship with post-reform government spending per capita (g).

Relative Tax Burden / Meltzer–Richard. To test the Meltzer–Richard hypothesis, I construct a measure of income inequality derived from household spending on property. In Meltzer and Richard (1981), as well as in the simple model in Section 2, government spending is determined by differences in the relative tax burden between the mean and median income earners, implying that the difference between mean and median income ratio is a predictor of redistributive pressure. In the historical context I study, taxes were levied not on income but on the rental value of occupied property—that is, the value at which a property could be rented. As such, as shown formally in Appendix A, the theoretical predictions in this setting relate to the ratio of mean to median household spending on property. I estimate this ratio as follows:

$$meanMedianRatio = \frac{Mean\ rental\ value}{Median\ wage} \tag{2}$$

The gross rental value for each district is available from the *Local Taxation Returns* and the median wage is estimated using county-level data on the 1869 agricultural wage from Collins and Thirsk (2000, Table 42.3). The median wage proxies for the median spending on property if the median voter spends the same proportion of their income on property across unions. The measure in (2) captures income inequality if spending on property increases monotonically with income (i.e., it is a normal good).[17]

In Appendix D.1, I present a number of robustness tests to ensure that the results are not driven by measurement assumptions. First, I replace the rental value in the numerator of the inequality measure with a direct measure of the tax base (the "rateable value"). Second, I replace the denominator with wage data from 1892 (county-level) and 1894 (union-level), to test sensitivity to the choice of year and the geographic unit.[18] Third, I use the HISCAM occupational score to proxy the status of the median household without relying on wage data. The main results are robust to using these alternative inequality measures.

The mean–median ratio is particularly valuable because, in contrast to other inequality measures, it directly captures the share of the tax burden borne by the poor—the mechanism at the core of the Meltzer–Richard hypothesis. As shown in Appendix Table B.3, a higher mean–median ratio is associated with several of the other inequality measures discussed below, including inequality in land ownership, the share of agricultural laborers, and the share of large farms. This demonstrates that the mean–median ratio captures some of the variation in these other measures, as we would expect given that they also reflect aspects of the income distribution. However, in this setting, these alternative measures are only indirectly related to the relative tax burden—a large landowner would not, for example, pay tax if they rented out their land. Consequently, they provide a less precise test of the Meltzer–Richard channel. As such, I use them to test the broader hypothesis that inequality shapes post-reform government spending. Further, including these other variables alongside the mean–median ratio in a regression allows me to separate the importance of the relative tax burden from these other theoretical channels.

Elite De Facto Power. Even if higher inequality generates redistributive pressure, it may not translate into higher spending if the wealthy retain effective control over policy (the welfare weight on the poor, φ, is low). I capture potential elite power using two measures. National political influence is measured using a binary indicator for whether a peer of the realm had a seat within the union. Local political control is

17. Available evidence supports these assumptions. First, the agricultural wage reflected not only the wage paid to workers, but also provides a guide to local wages more generally (Hunt 1973). Second, for typical agricultural households the income elasticity of housing appears to have been close to one: Horrell (1996) reports that households in low-wage agricultural regions (1840–1854) spent 10.4% of income on housing, compared to 9.9% in high-wage regions. This small variation supports the assumption that the median wage is highly correlated with median rental expenditure. Third, at higher income levels, income elasticity of housing was likely less than one, but still positive: the 1852 Select Committee on Income Taxation (Qs 5383) estimated that the share of income spent on rent declined from around one-eighth to one-tenth between annual incomes of £160 and £1,500 per annum—well above the typical agricultural wage (generally under £50). Similarly, Engels' original estimates (reported in Chai and Moneta 2010) suggest an income elasticity of housing of around 1 at low income levels, falling to approximately 0.7 at higher levels.

18. The 1892 and 1894 wage data provide a less precise measure of income and are available for fewer unions than the 1869 measure. Using the 1869 inequality measure also avoids potential endogeneity if inequality itself reacted to political changes in the 1890s.

measured using a binary indicator for whether the pre-reform chairman of the board of guardians was an unelected (appointed *ex officio*) member of the local gentry. Because the chairman was elected by the board members, this indicator identifies whether the local elite commanded the support of other guardians.

Fairness Concerns / Inequality Aversion. Inequality may also increase pressure for redistribution due to fairness concerns, such as an inherent aversion to inequality—summarized by the parameter κ in the model. To investigate this channel empirically, I use three measures that may affect the salience of differences between classes in this particular setting.[19] To capture the importance of wealthy landowners I use land inequality, estimated as the share of land owned by the very wealthy in each union. High land concentration may increase the salience of economic differences between classes, generate resentment, or skew understandings of the income distribution, and associated tax burdens. This mechanism could confound the test of the Meltzer–Richard mechanism, as a high mean–median ratio could reflect the presence of a small number of very salient large landowners, leading to a high tax for reasons unrelated to the tax base. Second, I use the share of large farms (those with more than five employees), which could increase the salience of agrarian class inequality. Third, I directly measure a source of grievances between classes—the harshness of policy during the "Crusade Against Outrelief". A harsher crusade could generate resentment as the poor felt the landed elites had reneged on their paternalistic responsibilities (Hurren 2015). I measure harshness as the percentage decline in the number of outdoor paupers between pre-crusade (1866–70) and post-crusade (1881–85) years.

Economic Need. The final row of the table reflects the fact that the existence of economically vulnerable groups may drive government spending. In the model, this is captured by the probability of being unemployed $(1-\pi)$ and the share of the population that is poor (λ). These groups have low incomes, and hence indirectly contribute to income inequality. However, the theoretical prediction regarding the effect of greater need for poor relief on spending is ambiguous. On one hand, more recipients mechanically increase total relief spending; on the other, they raise the marginal financial cost of more generous relief. Economic need could, in principle, also affect spending through other channels such as altruism (operating through κ) or through increasing the number of dependents in a household and hence relief received by households in employment (ψ).

I use three measures to capture variation in economic need. First, higher suitability for cereal production is expected to be correlated with a higher proportion of highly-seasonal cereal agriculture and hence a need for poor relief to support households outside of harvest season relative to areas dominated by pastoral agriculture. Second, I use census data to measure the share of male rural laborers amongst the population employed in agriculture. Third, the percentage of the population over 64 identifies

19. Agricultural economies often featured stark hierarchies, with large landowners and economically dependent landless laborers. Such conditions can fuel resentment and demands for redistribution (Domènech and Sánchez-Cuenca 2022), particularly if inequality is visible, or landowners are perceived to have broken a paternalistic social contract (Scott 1977).

areas with a high number of elderly citizens likely to rely on poor relief prior to the advent of a national pension system.

5.3. Inequality and Agriculture Type

The data show a clear pattern whereby areas with high cereal suitability—and hence arable agriculture—tend to be more unequal, as shown in Figure 3. There is clear regional clustering according to the level of income inequality, with more unequal areas generally used for higher value crops: income inequality is positively correlated with the 1885 percentage of county land used for wheat (r= 0.15, p-val=0.03) and negatively correlated with that used for oats (r= −0.31, p-value=0.00). Cereal suitability is strongly correlated with all the inequality measures except for local elite de facto power (see Appendix B.3). This pattern may be explained by the fact that higher value agricultural production affects patterns of land settlement and land value, and hence inequality (Cinnirella and Hornung 2016). Cereal agriculture also requires more farm labor than pastoral agriculture, leading to a larger mass of rural laborers.

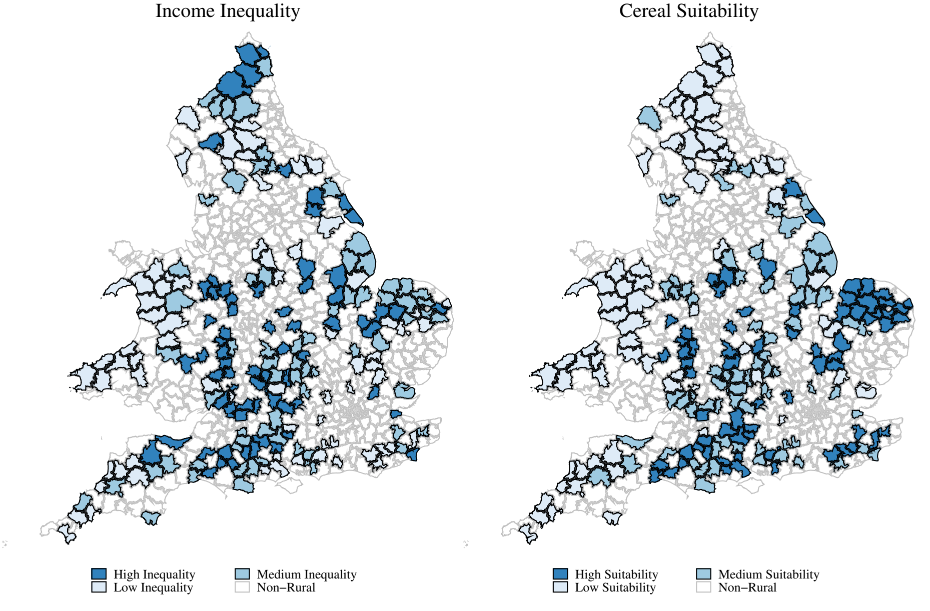

FIGURE 3. Poor law unions with high inequality are geographically clustered. The left-hand panel of the figure displays the estimated mean–median ratio from Equation (2). The right-hand panel displays FAO soil cereal suitability. High, medium, and low refer to terciles.

The connection between agriculture type and inequality poses two challenges for the empirical analysis. First, as discussed above, workers in arable agriculture were more reliant on poor relief to support their families outside harvest season.[20] As a result, it is important to test whether other inequality measures represent an independent causal channel, or instead simply capture greater need for poor relief in cereal-producing unions. Second, during this period cereal production in Britain declined rapidly due to the "grain invasion" from Eastern Europe and the United States—an economic shock that could have created growing demand for poor relief in more unequal areas, threatening the conditional parallel trends assumption. While the main impact of the shock was before the analysis period, I account for this concern by controlling for agriculture type in the main specifications and adding robustness tests allowing for different time trends according to land type.

Figure 4 provides evidence that differences in union characteristics are not a concern in practice. Despite the clear correlation between income inequality and various other variables across unions (left-hand panel), inequality does *not* appear correlated with the changes in observable characteristics pre-reform. This result suggests that agricultural type did not lead to changing demand for poor relief in the years leading up to the reform, supporting the validity of the identifying assumption. A more formal examination follows in Section 6.3.

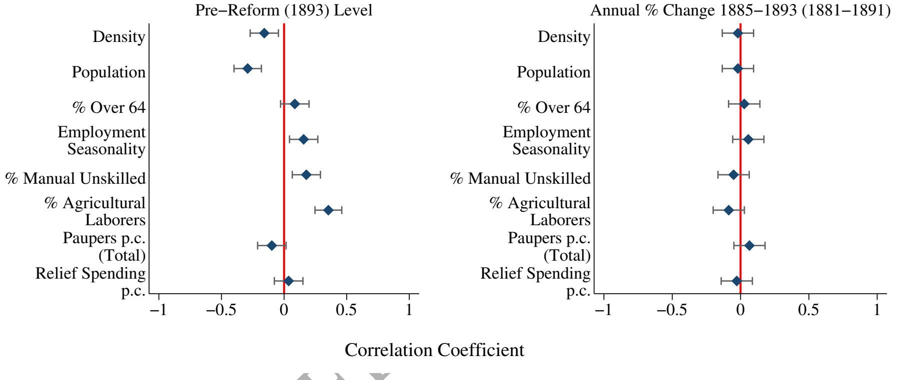

FIGURE 4. Income inequality is not correlated with trends in other union characteristics. The figure displays regression coefficient for a one standard deviation increase in income inequality on the level in 1893 (left-hand panel) or change 1885–1893 (right-hand panel). Bars represent 90% confidence intervals. For variables reported in the decennial census, levels are from 1891 and the change is for 1881–1891.

The geographic clustering of inequality also raises the concern that spatial autocorrelation could lead to spurious regression results—an issue I discuss in detail in Appendix D.4. First, as recommended by Conley and Kelly (2025), I calculate Moran's I (Moran 1948) for both the inequality measure and the change in relief spending. There is little evidence of spatial autocorrelation after accounting for agriculture type and the other regression controls. Second, I re-estimate the main regressions allowing

20. Seasonality in the number of outdoor paupers was correlated with the cereal suitability of a union ($r = 0.17$; p-value = 0.00).

for arbitrary spatial correlation between unions for a range of bandwidths (Conley 1999). The estimated standard errors are similar to standard errors clustered by poor law union, suggesting that spatial dependence is not inflating estimated precision. As a conservative measure, I use standard errors allowing for spatial autocorrelation within a bandwidth of 100 kilometers in the regression analysis. Finally, Table 2 demonstrates that the results are robust to allowing for flexible time trends according to location.

6. Results

The empirical analysis provides strong and robust evidence that more unequal areas experienced greater increases in government spending following the 1894 democratic reform. The first subsection demonstrates strong support for the Meltzer–Richard hypothesis. The second subsection tests all the theoretical hypotheses presented in Table 1, disentangling the effects of different types of inequality. The third subsection investigates the dynamic effects of inequality post-reform. The final subsection then disaggregates the effects of inequality on different components of poor relief expenditure.

6.1. Support for the Meltzer–Richard Hypothesis

Democratic reform led to larger increases in poor relief expenditure in areas with higher income inequality, as shown in Table 2. All variables are standardized, so that the regression coefficients represent the effect of a one standard deviation increase in each explanatory variable in terms of standard deviations of the outcome variable—per capita expenditure on poor relief. The coefficient on income inequality is positive and statistically significant across all specifications—areas with higher income inequality experienced greater increases in spending following the Local Government Act, consistent with the Meltzer–Richard hypothesis.

Importantly, the estimated coefficient on income inequality remains robust after including various sets of control variables. The coefficient remains stable across specifications after adding vectors of “crop controls” in (2), demographic controls (3), and financial controls (4). Specifications (5)–(7) allow flexibly for potential confounds by including quartic time trends interacted with union characteristics observed before the analysis period. First, to allow for trends in the demand for poor relief to differ according to agriculture type, I include a trend interacted with the suitability for cereal agriculture. In the second specification I allow for different time paths according to the ease of outmigration, proxied by the size of the nearest large town (over 25,000 population). Specification (7) allows spending of poor relief to evolve based on location, accounting for possible spatial trends in poor relief. Appendix Table D.5 shows similar specifications allowing for time trends according to the 1869 wage, population density, tax base per acre, the fall in tax base per acre 1875–1885, the distance from a major city, and seasonality of agriculture. The estimated coefficient on income inequality is of a similar magnitude and statistically significant across all specifications.

Table 2. High inequality districts had greater increases in expenditure following 1894 democratic reforms.

	DV=Relief Expenditure per Capita (Standardized)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Income Inequality x post1894	0.12***	0.09***	0.11***	0.11***	0.11***	0.11***	0.11***
	(.030)	(.028)	(.027)	(.026)	(.026)	(.027)	(.026)
Population			0.35	0.30	0.25	0.32	0.18
			(.453)	(.460)	(.463)	(.464)	(.474)
Population Density			−0.20	−0.26	−0.25	−0.26	−0.24
			(.435)	(.446)	(.447)	(.447)	(.448)
Decadal Variance in Pauperism			0.02	0.02	0.02	0.02	0.03
			(.021)	(.020)	(.019)	(.020)	(.019)
% Popn Male Age Over 64			0.02	0.02	0.00	0.02	0.01
			(.070)	(.068)	(.068)	(.068)	(.068)
% Popn Female Age Over 64			0.34***	0.32***	0.30***	0.32***	0.26***
			(.092)	(.090)	(.090)	(.090)	(.091)
% Revenue from Poor Rate				0.13***	0.12***	0.13***	0.12***
				(.026)	(.026)	(.026)	(.025)
Log Tax Base Per Acre				−0.11	−0.08	−0.11	−0.03
				(.114)	(.117)	(.116)	(.117)
Estimated Effect of Inequality as % of pre-Reform Spending:							
10th Percentile	6%	5%	6%	6%	5%	6%	6%
Median	10%	8%	9%	9%	9%	9%	9%
90th Percentile	14%	11%	13%	13%	12%	13%	13%
Average	10%	8%	9%	9%	9%	9%	9%
% Δ Spend post-1894	70%	56%	63%	63%	63%	63%	63%
Crop Controls	N	Y	Y	Y	Y	Y	Y
Demographic Controls	N	N	Y	Y	Y	Y	Y
Quartic Time Trend					% Wheat 1885	Nearest Town Size	Longitude &Latitude
Year Fixed Effects	Y	Y	Y	Y	Y	Y	Y
PLU Fixed Effects	Y	Y	Y	Y	Y	Y	Y
No. Observations	4,368	4,368	4,368	4,368	4,368	4,368	4,368
No. PLUs	208	208	208	208	208	208	208

Notes: $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All variables are standardized. "Crop Controls" is the county-level percentage of land used for different crops. "Demographic Controls" include the percentage of population of each sex in different age groups. "Quartic time trend" is a quartic polynomial in time interacted with the variable named in that row. Nearest Town Size is the log population of the nearest town with more than 25,000 inhabitants in 1891. The first four rows in the second panel report the estimated effect of different levels of inequality relative to spending in 1893. "% Δ Spend post-1894" is the estimated average effect of inequality post-reform relative to the actual change in spending after 1894. Standard errors are corrected for temporal and spatial autocorrelation within 100km using the approach of Colella et al. (2019) and are reported in parentheses.

The second panel of Table 2 investigates the magnitude of the effect of higher income inequality, both in absolute terms and relative to the overall growth in spending after 1894. The estimated effects are modest relative to pre-reform spending levels. At the median level of inequality in the sample, the estimated effect is approximately a 9% increase in per capita spending relative to spending in 1893. At the 90th percentile, the estimated effect is around 13%. These muted effects align with the limited impact of democratization we have seen in Figure 1. Overall, per capita spending increased by just 15% in the decade following the reform, likely reflecting the remaining barriers to political participation of the poor discussed in Section 3.3. The estimates in the final row of the panel indicate that more than half of this growth was due to the presence of inequality: if unions had been entirely equal, increases in expenditure would have been markedly lower. Income inequality thus appears to have exerted an important influence on the—relatively small—effect of democratic reform.[21]

6.2. *The Effect of Alternative Dimensions of Inequality*

I now investigate other potential mechanisms linking inequality and government spending listed in Table 1. I first re-estimate specification (4) from Table 2 using each inequality measure separately, and then include multiple measures together to disentangle different channels. To allow comparisons across specifications, each non-binary inequality measure is again standardized.

The results, presented in Table 3, provide some evidence for all four causal channels. When examining each hypothesis independently, I find (sometimes weakly) statistically significant coefficients in the expected direction for all measures except the presence of an aristocrat, which proxies national political influence. The analysis thus provides some support for various theoretical channels suggested in previous studies, with elite de facto reflecting local rather than national influence.

The final two columns of Table 3 suggest that the Meltzer–Richard model identifies a distinct mechanism through which inequality affects government spending. The coefficient on income inequality remains similar in size and statistical significance when controlling for all the other inequality measures. These specifications thus provide reassurance that the mean–median rental ratio—and hence the relative tax burden—captures an effect of inequality that is distinguishable from local economic structure and other inequality measures.

There is also consistent evidence that spending grew more in arable-intensive unions and less in unions controlled by local elites. Elite de facto power $(1 - \varphi)$ thus appears to have played an important role in shaping the effect of democratic reform. In contrast, the coefficients relating to fairness concerns (κ) shrink considerably and

21. As a comparison, Husted and Kenny (1997) find that eliminating each of poll taxes and literacy tests in the United States in the 1960s raised state welfare spending by between 10% and 20%. Although these estimates are not directly comparable, they suggest that the reforms here had a smaller effect despite starting from a lower pre-reform base level of taxation, and involving a reform that had much broader electoral consequences.

become statistically indistinguishable from zero once cereal suitability is controlled for. The apparent effects of these variables may thus capture other, correlated, local characteristics—particularly economic structure—rather than indicating an independent causal mechanism. Alternatively, cereal suitability may capture multiple mechanisms through which inequality affected government spending, and our measures are insufficiently nuanced to tease them apart.

Appendix D.1 examines alternative operationalizations of the different dimensions of inequality. This analysis uses alternative definitions of some measures—the percent of wheat or percent of old-aged men or women to capture the need for poor relief, and the drop in adult paupers to capture harsh relief policies. For other inequality measures I introduce alternative variables entirely. Specifically, as alternative sources of fairness concerns I use land inequality in terms of land value (not size), wealth inequality (estimated using the number of servants), and the number of gentry in a union. As an alternative to aristocratic presence, I use the presence of a great landowner in case wealth, rather than national prestige, provided de facto power. The conclusion that the effect of democratic reform was shaped by income inequality, economic structure, and elite de facto power is robust to these alternative specifications. Further, the results are similar when allowing for alternative definitions of the mean–median income ratio—see Appendix Table D.1.

Appendix D reports a number of additional robustness tests. First, I allow for the effect of the reform to vary according to a range of characteristics that could affect the demand for relief, including both variables to capture differences in economic need (land value, wages, and the seasonality of pre-reform relief; Appendix Table D.3) and the ease of outmigration (distance to nearby towns, population density, and change in demographic structure; Appendix Table D.4). The coefficients on these variables are sometimes statistically distinguishable from zero, but they do not disturb the main findings regarding inequality. Second, I allow for different time trends according to the observable characteristics of different poor law unions—a further check that omitted variable bias does not explain the results (Appendix Table D.5). Third, I show that the results are not driven by particular sub-samples of poor law unions (Appendix Table D.6). Fourth, I account for differences in urban surroundings, which could plausibly drive migration and wage dynamics. In particular, I control for the population, population growth, and tax base of the nearest town to each union (Appendix Table D.7), as well as allowing for the reaction of the reform to differ according to these characteristics (Appendix Table D.8). The magnitude and statistical significance of the coefficients relating to inequality remains similar across these specifications.

In summary, the analysis provides robust evidence that the effect of democratic reform on spending was shaped by three dimensions of inequality. Spending increased more in areas of higher income inequality, consistent with the Meltzer–Richard model. It also grew more in unions with high cereal suitability, which may reflect greater need for income support. In contrast, spending grew less where elites controlled poor law boards before reform. However, after accounting for differences in economic structure, there is little evidence that fairness concerns played a significant role in shaping post-reform spending. The findings thus provide evidence for both median voter models of redistribution and theories emphasizing the importance of elite de facto power.

6.3. Dynamic Effects

This subsection investigates the dynamic effect of the democratic reform, focusing on the three forms of inequality—income inequality, elite de facto power, and economic need—which emerged as important previously. To do so I estimate flexible regression specifications allowing the effect of each inequality measure to differ every year. Doing so both investigates the timing of the effects identified in Table 3—whether they emerged immediately post reform, and whether they persisted over time—and also checks whether there is any evidence of pre-trends before 1894 that would suggest the identifying assumptions are violated. This set-up also allows a more stringent test of causal identification through the inclusion of union-specific linear time trends that capture any pre-reform trends in spending.

The dynamic analysis, presented in Figure 5, provides further evidence that spending increased in areas of high income inequality (left-hand side) and those of higher economic need (central column). There is little evidence of any relationship between either variable and spending pre-reform—supporting the conditional parallel trends assumption—but a clearly increasing trend immediately after. Notably, areas with more cereal suitability experienced an upward jump in spending after 1894, consistent with this variable capturing the presence of needy families that would immediately benefit from more generous provision. Further, the bottom panel of Figure 5 demonstrates robustness to the inclusion of union-specific linear time trends.[22]

We see a different pattern when examining elites' de facto control. Areas with ex officio chairmen appear to have had falling spending even before the 1894 reform, particularly before 1890. Such a result makes sense both historically, given the Crusade Against Outrelief, and theoretically—if elites had control, why would they not use it? However, this pattern suggests that the results in Table 3 may partly reflect elite efforts to reduce spending pre-reform. Nevertheless, Figure 5 indicates that democratization strengthened elites' attempts to reduce expenditure.

22. These specifications allow for interactions between inequality and the year dummies only post-reform, so that the union-specific trends do not capture part of the dynamic response to the reform (see Wolfers 2006). In the dynamic specifications, I use an 80 km bandwidth to account for spatial autocorrelation, ensuring sufficient spatial variation to estimate standard errors for the individual year coefficients.

TABLE 3. Effect of democratic reform was conditioned by multiple dimensions of inequality.

	DV=Relief Expenditure per Capita (Standardized)										
	Meltzer-Richard ($\bar{y}/y_P$)	Elite De Facto Power ($1-\varphi$)		Fairness Concerns (κ)			Economic Need (π, λ)			Multiple Channels	
Post 1894 x											
Income Inequality	0.11***									0.09***	0.10***
	(.026)									(.028)	(.026)
Local Elite Control		−0.13**								−0.11**	−0.11**
		(.057)								(.053)	(.054)
Aristocratic Elite			−0.00							−0.00	
			(.054)							(.052)	
Land Inequality				0.05*						0.03	
				(.026)						(.025)	
% Large Farms					0.06**					−0.04	
					(.029)					(.038)	
Outrelief Cuts 1866–85						0.06*				0.02	
						(.031)				(.030)	
Cereal Suitability							0.11***			0.08**	0.10***
							(.031)			(.035)	(.030)
% Agricultural Laborers								0.10***		0.04	
								(.026)		(.039)	
% Age over 64									0.08**	0.06*	
									(.032)	(.030)	
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
PLU Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
No. Observations	4,368	4,368	4,368	4,368	4,368	4,368	4,368	4,368	4,368	4,368	4,368
No. PLUs	208	208	208	208	208	208	208	208	208	208	208

Notes: All non-binary variables are standardized. "Controls" include the variables included in specification (4) of Table 2. Standard errors are corrected for temporal and spatial autocorrelation within 100km using the approach of Colella et al. (2019) and are reported in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

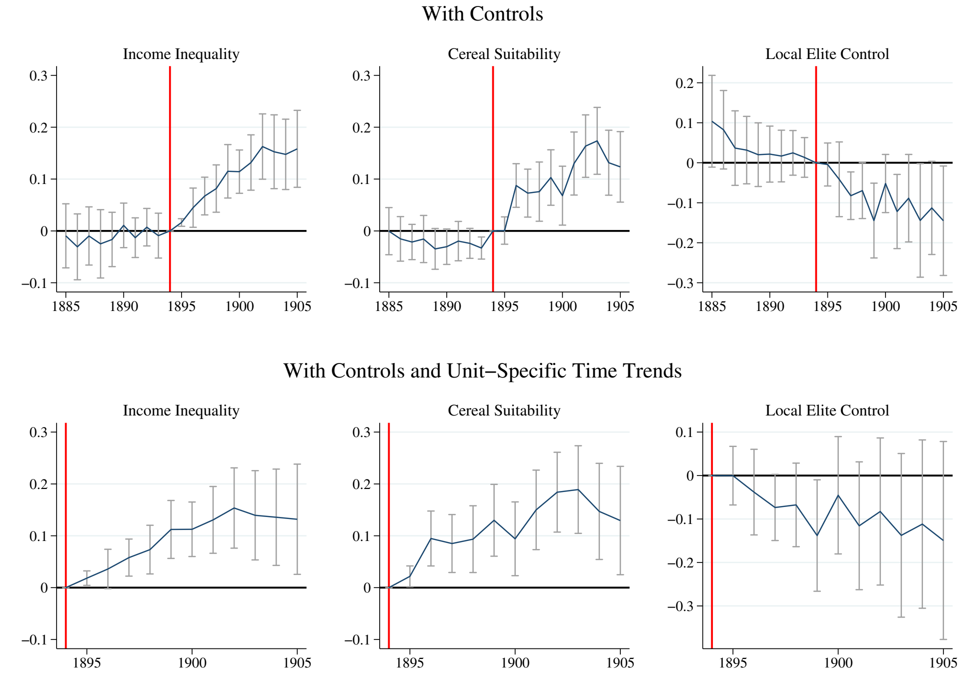

FIGURE 5. The relationship between inequality and relief spending emerges only after the 1894 democratic reform. The top panels of the figure plot β_j and β_k, and associated 90% confidence intervals, from the specification:

$$y_{it} = \sum_{j<1894} \beta_j \left(Inequality_i \times year_j\right) + \sum_{k\geq 1895} \beta_k \left(Inequality_i \times year_k\right) + \delta X_{it} + \alpha_i + \mu_t + \varepsilon_{it}$$

where *Inequality* refers to the variable in the title of each panel. Similarly, the bottom panels plot γ_k and associated 90% confidence intervals, from the specification:

$$y_{it} = \sum_{k\geq 1895} \gamma_k \left(Inequality_i \times year_k\right) + \delta X_{it} + \alpha_i + \mu_t + T_{it} + \varepsilon_{it}$$

where T_{it} are union-specific linear time trends. The excluded category in both cases relates to 1894, so results are relative to the year of reform. All specifications include controls from specification (4) of Table 2. Standard errors are corrected for temporal and spatial autocorrelation within 80km using the approach of Colella et al. (2019).

6.4. Disaggregating the Effects on Poor Relief Expenditure

Table 4 investigates the ways in which inequality shaped poor relief expenditure. I investigate both different spending categories and also the number of recipients of relief. In both cases I distinguish between relief provided outside of the workhouse ("outdoor")—at the heart of many political debates about poor relief—and in the workhouse ("indoor"). I also examine spending, such as salaries of poor law officers, that was not directly related to pauper maintenance.

The results suggest that the effect of higher income inequality was predominantly on spending on outdoor relief, consistent with both the theoretical predictions and historic evidence. Conceptually, we can consider the provision of outrelief as capturing ψ, the fraction of relief received by households while employed. The minimal evidence of any effect of income inequality on spending on indoor relief is consistent with the Meltzer–Richard prediction only applying if ψ is sufficiently high. Contemporary evaluations of the Act, summarized in Appendix C.2, also emphasized the effect of the reform on outrelief. In Stow Union, for example, prior to 1894 the guardians "were very particular as to the grant of outrelief, and if the least doubt existed as to the condition of the applicant, an order for admission to the workhouse only was given. In the year 1895, the new qualification admitted a majority of laborers to the board, and [their] policy was immediately subverted...outrelief jumped from £45 to £92 a week, and later to £100."

TABLE 4. The primary effect of income inequality was increasing outdoor relief.

	Dependent Variable (Standardized)					
	Expenditure p.c.			Paupers p.c.		
	Outdoor	Indoor	Other	Total	Outdoor	Indoor
Post 1894 x						
Income Inequality	0.09***	0.00	0.06**	0.11***	0.08***	0.04
	(.027)	(.033)	(.027)	(.027)	(.022)	(.025)
Local Elite Control	0.00	−0.08	−0.15***	0.02	0.02	−0.08
	(.058)	(.067)	(.054)	(.055)	(.050)	(.051)
Cereal Suitability	0.03	0.06*	0.11***	0.02	0.03	0.03
	(.028)	(.037)	(.033)	(.029)	(.026)	(.027)
Controls	Y	Y	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y	Y	Y
PLU Fixed Effects	Y	Y	Y	Y	Y	Y
No. Observations	4,368	4,368	4,368	4,368	4,366	4,364
No. PLUs	208	208	208	208	208	208

Notes: "Controls" include the variables included in specification (4) of Table 2. Standard errors are corrected for temporal and spatial autocorrelation within 100km using the approach of Colella et al. (2019) and are reported in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

The results relating to the other inequality measures are also compatible with the theoretical framework. The negative effect of elite control is focused on other expenditure and, more weakly, spending in the workhouse. This pattern suggests that reform may have forced elites to be more generous in publicly-salient—and hence high φ—elements of relief after 1894, while using their political influence to reduce less observable forms of expenditure. The impact of greater economic need is clearest on other expenditure, which is consistent with scaling up the overall provision of relief, rather than expanding the generosity of relief for each recipient. The overall pattern is thus consistent with the model in Section 2, although it is important to note that the mapping from the theoretical parameters to these disaggregated outcomes is not clear-cut, and partly reflects ex post interpretation.

Further analysis, presented in Appendix D.5, provides suggestive evidence that the main effect of the 1894 reforms was to increase support for the elderly and women, rather than the adult male workforce. Appendix Table D.9 estimates the effect of the reforms on different categories of paupers, classified by age (adult versus child), sex, and ability to work ("able-bodied" versus "non-able-bodied," a category that in practice included many elderly paupers). These latter categories should be treated with some caution as they were not well-defined and likely manipulable to justify the provision of relief. Income inequality leads to a clear increase in the number of non-able-bodied outdoor paupers as well as outdoor able-bodied adult women, but there is no clear effect on indoor pauper numbers, or on able-bodied men. Higher economic need (cereal suitability) is associated with greater support to non-able-bodied paupers, suggesting that democratic reform did not expand the use of poor relief as a means of insuring against seasonal unemployment.

7. Conclusion

This paper provides empirical support for Meltzer and Richard (1981)'s prediction that higher income inequality leads to greater government spending on redistribution. I test the effects of democratization on government redistributive spending in a setting that closely aligns with the assumptions underpinning the Meltzer–Richard model—autonomous, locally-elected councils funding welfare through an approximately proportional tax levied on all households. This institutional context allows me to isolate the core theoretical mechanism—the relative tax burden borne by rich and poor—while sidestepping many confounding factors present in national-level analyses. The results demonstrate that national democratic reform in 1894 led to greater increases in spending in areas of higher pre-reform income inequality, consistent with theoretical predictions.

These findings illustrate the importance of separating the precise predictions of the Meltzer–Richard model from a more general hypothesis that inequality will increase government spending. Common measures of inequality, such as the Gini coefficient, may be poorly aligned with actual tax burdens and therefore weakly capture this mechanism. Indeed, in this paper, the estimated mean–median income ratio predicts post-reform spending more strongly than alternative inequality measures. Such empirical challenges may explain why many previous studies have found weaker evidence in support of the Meltzer–Richard hypothesis.

The results also support theories emphasizing the importance of inequality in shaping the results of democratization. Wealthy citizens had a greater incentive to resist democratization of the bodies controlling spending on poor relief in areas of high inequality. The results also add empirical support for elite de facto power dampening the effects of democratic reform. Moreover, a simple model suggests that Meltzer–Richard may be particularly relevant in settings where, as in nineteenth-century Britain, the poor have relatively little political power even after democratization.

The paper also sheds light on an understudied period in the history of the English Poor Law. The 1894 democratization of poor law boards marked a turning point in both the level and composition of expenditure, including a shift towards greater provision of relief outside the workhouse and, perhaps, increased support for the elderly. This finding contributes to our understanding of the nascent welfare state and the steps towards the introduction of national pensions in 1911. Further research is required to understand the broader consequences of the Local Government Act, and particularly whether it had similar effects on expenditure in urban areas. Those areas were also highly unequal, but may have exhibited different political dynamics. In particular, urban areas contained a lower middle class that was disinclined to support public spending (Lindert 2004), but also lacked the powerful landed elites that continued to constrain expenditure in rural areas after 1894.

Finally, the paper demonstrates how decentralization can constrain the redistributive impact of national democratization. Britain's 1832 Great Reform Act expanded the national franchise, but left the local political structures governing poor relief largely undisturbed (Lizzeri and Persico 2004). In fact, Parliament reformed these undemocratic bodies only after the 1884 Third Reform Act enfranchised poor rural households.[23] By securing control of local institutions in 1834, landed elites insulated themselves from redistributive pressures even as their national influence waned. Understanding when and how elites use decentralization to blunt the effects of democratic reform remains an important question for future research.

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23. The three national Reform Acts were each followed by reforms to local bodies in the areas where newly enfranchised voters were concentrated. The Great Reform Act enfranchised urban middle classes, and was followed by the restructuring of municipal councils (Lizzeri and Persico 2004). The 1867 Second Reform Act extended voting rights to the urban lower middle classes and was followed by further changes to municipal governance. The 1884 Third Reform Act enfranchised agricultural laborers, and was followed by the creation of elected county councils in 1888 and the reforms discussed in this paper.

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